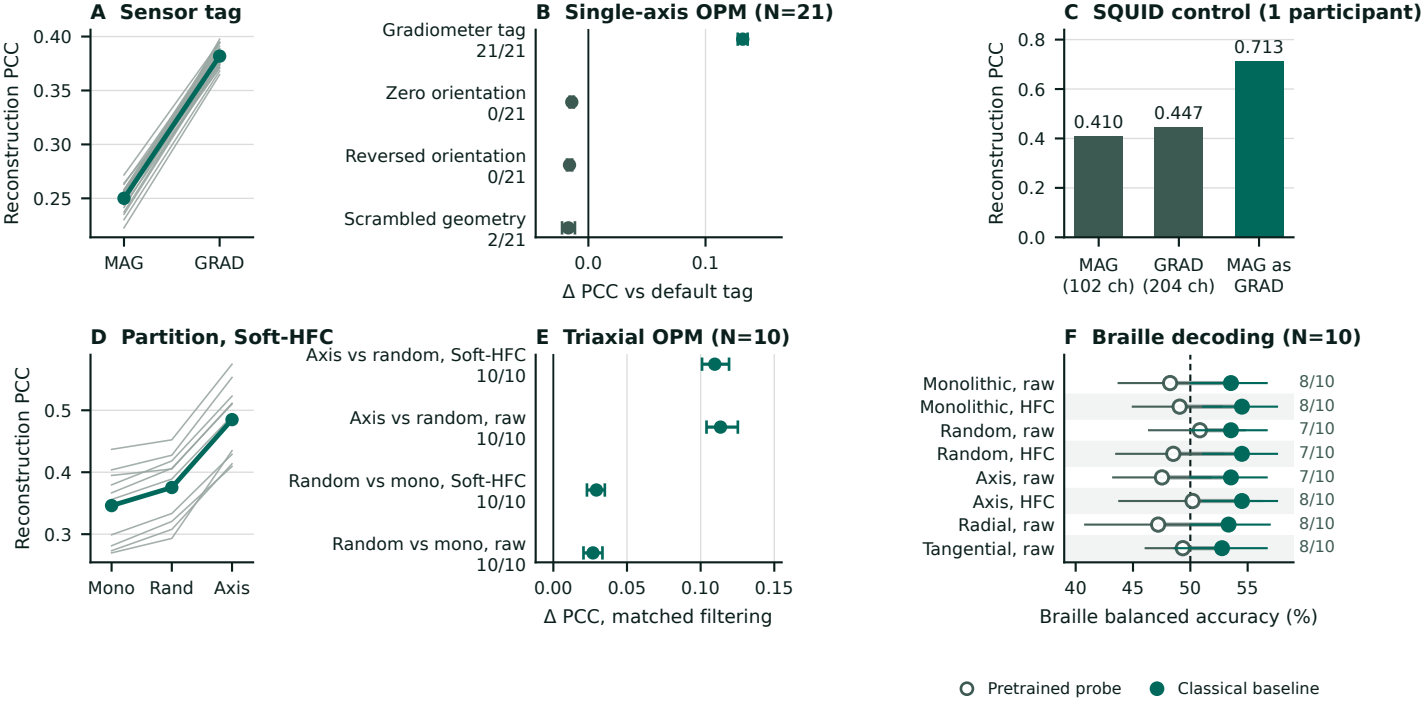
A Sensor tag
Reconstruction PCC
0.40
0.35
0.30
0.25
MAG
GRAD
B Single-axis OPM (N=21)
Gradiometer tag
21/21
Zero orientation
0/21
Reversed orientation
0/21
Scrambled geometry
2/21
0.0
0.1
Δ PCC vs default tag
C SQUID control (1 participant)
Reconstruction PCC
0.8
0.6
0.4
0.2
0.0
0.410
0.447
0.713
MAG
(102 ch)
GRAD
(204 ch)
MAG as
GRAD
D Partition, Soft-HFC
Reconstruction PCC
0.5
0.4
0.3
Mono
Rand
Axis
E Triaxial OPM (N=10)
Axis vs random, Soft-HFC
10/10
Axis vs random, raw
10/10
Random vs mono, Soft-HFC
10/10
Random vs mono, raw
10/10
0.00
0.05
0.10
0.15
Δ PCC, matched filtering
F Braille decoding (N=10)
Monolithic, raw
Monolithic, HFC
Random, raw
Random, HFC
Axis, raw
Axis, HFC
Radial, raw
Tangential, raw
8/10
8/10
7/10
7/10
7/10
8/10
8/10
8/10
40
45
50
55
Braille balanced accuracy (%)
Pretrained probe
Classical baseline